

Leetcode 787: Cheapest Flights Within K stops.

Experiment - 7

|| Dry Run ||

// Dry run for example.

flights = {

{0, 1, 50},

{0, 2, 300},

{1, 2, 50},

{2, 3, 100},

{1, 3, 200},

{0, 3, 1000}

};

n = 4, src = 0, dest = 3, k = 2

* flights = { source, destination, cost to reach }

* n $\Rightarrow$ number of nodes/cities.

* src $\Rightarrow$ Starting vertex.

* dest $\Rightarrow$ Goal vertex.

So in this question we have to find cost (minimum) to reach 3 from 0.

|| Adjacency list ||

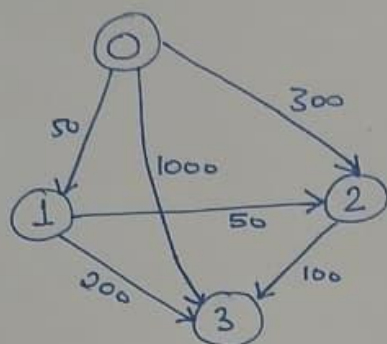
0 $\rightarrow$ [(1, 50), (2, 300), (3, 1000)]

1 $\rightarrow$ [(2, 50), (3, 200)]

2 $\rightarrow$ [(3, 100)]

3 $\rightarrow$ []

Goal: * To reach 3 from 0 with minimum cost
 * Max 2 stops are allowed.



Initial state

$$\text{Cost} = [0, \infty, \infty, \infty]$$

$$q = [(0 \text{ stops}, \text{node}=0, \text{cost}=0)]$$

Step 1:

$$\text{node} \rightarrow 0, \text{ stops} \rightarrow 0$$

Neighbour:

$$\begin{array}{l} 0 \rightarrow 1 (50) \\ 0 \rightarrow 2 (300) \\ 0 \rightarrow 3 (1000) \end{array}$$

$$\text{Queue} = [(1, 1, 50), (1, 2, 300), (1, 3, 1000)]$$

Step 2:

$$\text{node} \rightarrow 1, \text{ stops} \rightarrow 1$$

neighbour:

$$\begin{array}{l} 1 \rightarrow 2 (50) \rightarrow \text{new cost} = 100 < 300 (0 \rightarrow 2) \\ 1 \rightarrow 3 (200) \rightarrow \text{new cost} = 50 + 200 = 250 < 1000 (0 \rightarrow 3) \end{array}$$

$$\text{Cost}[2] = 100$$

$$\text{Cost}[3] = 250$$

$$\text{Queue} = [(1, 2, 300), (1, 3, 1000), (2, 2, 100), (2, 3, 250)]$$

Step 3: node $\rightarrow 2$, stop $\rightarrow 1$, cost $\rightarrow 300$.

- $2 \rightarrow 3$ (100) $\rightarrow$ new cost = 400 $>$ 250 X
No update.

Step 4: node $\rightarrow 3$, stop $\rightarrow 1$.
No edge left.

Step 5: Node $\rightarrow 2$, stop $\rightarrow 2$, cost $\rightarrow 100$

- $2 \rightarrow 3$ (100) $\rightarrow$ new cost = 200 $<$ 250 $\checkmark$

$$\text{cost}[3] = 200$$

$$\text{Queue} = [(2, 3, 250) , (3, 3, 200)]$$

Step 6: Node $\rightarrow 3$ (stop 2, cost = 250)

$\rightarrow$ No update as cost $>$ 200.

Step 7: stop will be $\rightarrow 3$ X
ignore.

Final cost Array:

$$\text{cost} = [0, 50, 100, 200]$$

Ans $\rightarrow$ Minimum cost $\rightarrow$ $0 \rightarrow 3 = 200$
Best Path $\rightarrow 0 \rightarrow 1 \rightarrow 2 \rightarrow 3$

2 steps $\rightarrow 2$

Ans